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MATCH: Execution-Aware Hierarchical Reinforcement Learning for Vision-and-Language Navigation

Joint End-to-End Training of Planner and Executor Sets New State-of-the-Art on R2R-CE and RxR-CE at 0.5B Parameters

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Hierarchical VLN agents that optimize the high-level planner while keeping the low-level executor fixed suffer from execution blindness: the planner proposes waypoints without knowing whether the executor can reach them. MATCH closes this loop by jointly training a language-conditioned planner and a goal-conditioned executor end-to-end with reinforcement learning, routing execution feedback back to inform future waypoint selection. With only 0.5B parameters, MATCH sets new state-of-the-art on R2R-CE (62.4% SR) and RxR-CE (55.8% SR).

AT A GLANCE

Problem: In VLN-CE, hierarchical planners propose waypoints without knowing what the executor can achieve, causing cascading failures.

Why it matters: Embodied navigation in continuous 3D environments with language instructions is a core capability for household robots and assistive AI.

Method: End-to-end RL training of planner and executor; execution-status feedback from executor to planner at each step; shared visual backbone.

Result: 62.4% SR on R2R-CE (val unseen) and 55.8% SR on RxR-CE (val unseen), surpassing all prior methods.

Evidence: R2R-CE and RxR-CE benchmarks, val unseen and test splits; ablations over execution feedback and joint training.

The Big Picture

Vision-and-Language Navigation (VLN) tests an agent's ability to follow free-form instructions like "walk past the sofa, turn right at the kitchen, and stop in front of the bookshelf" in a photo-realistic 3D environment. In Continuous Environments (VLN-CE), the agent must produce raw motor actions—translation velocity and rotation—rather than selecting named waypoints on a graph, making the problem significantly harder.

Hierarchical approaches decompose VLN-CE into two sub-problems: a *planner* that parses the instruction and generates a sequence of subgoals (waypoints), and an *executor* that drives the robot from its current position to each waypoint. This decomposition is intuitive but introduces a structural mismatch: the planner has no direct signal about whether its proposed waypoints are achievable given the current room geometry and the executor's capabilities.

Why Existing Approaches Fall Short

Fixed executor, trained planner: Most prior hierarchical VLN-CE systems pretrain or fix the low-level executor (typically a model-predictive controller or a simple goal-conditioned policy) and focus all learning effort on the high-level planner. The planner therefore cannot receive gradient signal from execution failures and learns only from its own prediction errors.

Mismatched subgoal abstraction: Even when both components are trained, they are typically trained separately on different objectives (language-goal matching for the planner; reaching accuracy for the executor), meaning there is no shared objective that penalizes the planner for proposing unreachable waypoints.

No execution feedback: The planner selects its next waypoint based on the instruction and the current visual observation, but not on how the last waypoint was executed. Information about partial failures (executor reached 70% of the waypoint but collided) is discarded between planning steps.

Core Mechanism

Language-Conditioned Planner. Given the full instruction I and the current visual observation o_t, the planner $\pi_P(w_t \mid I, o_t, h_t^P)$ produces a waypoint w_t as a displacement vector relative to the current agent pose, where h_t^P is the planner's recurrent state.

Goal-Conditioned Executor. Given the current observation o_t^{low} and the target waypoint w_t, the executor $\pi_E(a_t \mid o_t^{\text{low}}, w_t, h_t^E)$ produces a primitive action a_t (linear velocity, angular velocity).

Execution-Status Feedback. After the executor attempts waypoint w_t, the achieved displacement and an obstacle-collision indicator are compressed into an execution-status vector s_t and appended to the planner's observation at step $t+1$: $h_{t+1}^P = \text{RNN}(h_t^P, [o_{t+1}, s_t])$.

Joint RL Training. The planner and executor share a single reward: the navigation success reward at the

episode terminus, discounted back through the waypoint sequence. Both are updated by PPO in a shared replay buffer.

Technical Formulation

The planner's policy is:

$$w_t = \pi_P\big(I,\, o_t,\, h_t^P,\, s_{t-1}\big)$$

The executor's policy unrolls for T_{exec} steps per waypoint:

$$a_\tau = \pi_E\big(o_\tau^{\text{low}},\, w_t - p_\tau,\, h_\tau^E\big)$$

where p_τ is the agent's current position.

The reward shaping for the planner uses a distance-based dense reward:

$$r_t^P = d(p_t, g) - d(p_{t+1}, g) - \lambda_c \cdot \mathbf{1}[\text{collision}_t]$$

where g is the goal location and λ_c penalizes collisions. The executor receives a waypoint-reaching reward:

$$r_\tau^E = \max(0,\; d(p_\tau, w_t) - d(p_{\tau+1}, w_t))$$

The full PPO objective optimizes π_P and π_E jointly:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}}(\pi_P) + \beta\, \mathcal{L}_{\text{PPO}}(\pi_E)$$

Learning or Inference Procedure

Pretraining. The shared visual backbone (ViT-B) is initialized from a pretrained visual-semantic embedding. The instruction encoder is initialized from a compact BERT variant.

Stage 1 – Imitation Learning. Both planner and executor are pretrained with behavior cloning on expert trajectories for 10k steps.

Stage 2 – Joint RL Fine-tuning. PPO is applied for 50k environment steps. The execution-feedback vector is activated after Stage 1, ensuring the planner does not receive noisy feedback from an untrained executor.

Inference. The planner generates one waypoint at a time, the executor runs for up to $T_{\text{exec}} = 20$ steps, and the planner then receives the execution-status feedback and produces the next waypoint.

What the Guarantee Says

No theoretical guarantee is claimed. The paper's argument is empirical: execution-aware training produces better navigation policies, and the ablation studies isolate which components contribute. The success-rate metric on val unseen splits, which feature environments not seen during training, provides evidence of generalization.

Experimental Findings

- **R2R-CE val unseen:** MATCH achieves **SR = 62.4%**, **GP = 65.1%**, **NE = 3.8 m**, surpassing the prior SOTA (SR = 58.1%) by 4.3 points.
- **RxR-CE val unseen:** MATCH achieves **SR = 55.8%**, improving over the prior best (51.3%) by 4.5 points.
- **Parameter efficiency:** MATCH achieves these results with **0.5B parameters**, outperforming prior methods that use 2–7B parameters.
- **Collision rate:** Joint RL training with the collision penalty reduces the executor's collision rate by 31% relative to the imitation-only baseline.

Ablations and Interpretation

Execution feedback removed: Removing the execution-status vector and training jointly on reward alone reduces SR on R2R-CE by 3.1 points, confirming that the information channel—not just joint training—is critical.

Fixed executor (standard hierarchical baseline): Fixing the executor at its imitation-learning checkpoint and training only the planner reduces SR by 5.7 points, establishing the cost of decoupled training.

Imitation only (no RL): Without RL, SR is 54.6%, 7.8 points below MATCH. RL is necessary to generalize to execution scenarios not covered by expert data.

Waypoint granularity: Decreasing the maximum number of waypoints per episode degrades SR smoothly, suggesting the planner learns to use waypoints efficiently rather than uniformly spacing them.

Reference

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